

MATHEMATICS

CLASS X

There will be **one** paper of **three** hours duration carrying 80 marks and Internal Assessment of 20 marks.

Certain questions may require the use of Mathematical tables (Logarithmic and Trigonometric tables).

1. Commercial Mathematics

(i) Goods and Services Tax (GST)

Computation of tax including problems involving discounts, list-price, profit, loss, basic/cost price including inverse cases. Candidates are also expected to find price paid by the consumer after paying State Goods and Service Tax (SGST) and Central Goods and Service Tax (CGST) - the different rates as in vogue on different types of items will be provided. Problems based on corresponding inverse cases are also included.

(ii) Banking

Recurring Deposit Accounts: computation of interest and maturity value using the formula:

$$I = P \frac{n(n+1)}{2 \times 12} \times \frac{r}{100}$$

$$MV = P \times n + I$$

(iii) Shares and Dividends

(a) Face/Nominal Value, Market Value, Dividend, Rate of Dividend, Premium.

(b) Formulae

- Income = number of shares $\times$ rate of dividend $\times$ FV.
- Return = (Income / Investment) $\times$ 100.

Note: Brokerage and fractional shares **not** included.

2. Algebra

(i) Linear Inequations

Linear Inequations in one unknown for $x \in N, W, Z, R$. Solving:

- Algebraically and writing the solution in set notation form.
- Representation of solution on the number line.

(ii) Quadratic Equations in one variable

(a) Nature of roots

- Two distinct real roots if $b^2 - 4ac > 0$
- Two equal real roots if $b^2 - 4ac = 0$
- No real roots if $b^2 - 4ac < 0$

(b) Solving Quadratic equations by:

- Factorisation
- Using Formula.

(c) Solving simple quadratic equation problems.

(iii) Ratio and Proportion

(a) Proportion, Continued proportion, mean proportion

(b) Componendo, dividendo, alternendo, invertendo properties and their combinations.

(c) Direct simple applications on proportions only.

(iv) Factorisation of polynomials:

(a) Factor Theorem.

(b) Remainder Theorem.

(c) Factorising a polynomial completely after obtaining one factor by factor theorem.

Note: $f(x)$ not to exceed degree 3.

(v) Matrices

(a) Order of a matrix. Row and column matrices.

(b) Compatibility for addition and multiplication.

(c) Null and Identity matrices.

(d) Addition and subtraction of 2×2 matrices.

- (e) *Multiplication of a 2×2 matrix by*
 - *a non-zero rational number*
 - *a matrix.*
 - (vi) *Arithmetic and Geometric Progression*
 - *Finding their General term.*
 - *Finding Sum of their first 'n' terms.*
 - *Simple Applications.*
 - (vii) *Co-ordinate Geometry*
 - (a) *Reflection*
 - (i) *Reflection of a point in a line:*
 $x=0, y=0, x=a, y=a$, the origin.
 - (ii) *Reflection of a point in the origin.*
 - (iii) *Invariant points.*
 - (b) *Co-ordinates expressed as (x,y), Section formula, Midpoint formula, Concept of slope, equation of a line, Various forms of straight lines.*
 - (i) *Section and Mid-point formula (Internal section only, co-ordinates of the centroid of a triangle included).*
 - (ii) *Equation of a line:*
 - *Slope –intercept form $y = mx + c$*
 - *Two- point form $(y-y_1) = m(x-x_1)$*
 - *Geometric understanding of 'm' as slope/ gradient/ $\tan \theta$ where θ is the angle the line makes with the positive direction of the x axis.*
 - *Geometric understanding of 'c' as the y-intercept/the ordinate of the point where the line intercepts the y axis/ the point on the line where $x=0$.*
 - *Conditions for two lines to be parallel or perpendicular.*
- Simple applications of all the above.*
- (ii) *Comparison with congruency, keyword being proportionality.*
 - (iii) *Three conditions: SSS, SAS, AA. Simple applications (proof not included).*
 - (iv) *Applications of Basic Proportionality Theorem.*
 - (v) *Areas of similar triangles are proportional to the squares of corresponding sides.*
 - (vi) *Direct applications based on the above including applications to maps and models.*
- (b) *Loci*

Loci: Definition, meaning, Theorems and constructions based on Loci.

 - (i) *The locus of a point at a fixed distance from a fixed point is a circle with the fixed point as centre and fixed distance as radius.*
 - (ii) *The locus of a point equidistant from two intersecting lines is the bisector of the angles between the lines.*
 - (iii) *The locus of a point equidistant from two given points is the perpendicular bisector of the line joining the points.*

Proofs not required.
- (c) *Circles*
 - (i) *Angle Properties*
 - *The angle that an arc of a circle subtends at the centre is double that which it subtends at any point on the remaining part of the circle.*
 - *Angles in the same segment of a circle are equal (without proof).*
 - *Angle in a semi-circle is a right angle.*
 - (ii) *Cyclic Properties:*
 - *Opposite angles of a cyclic quadrilateral are supplementary.*
 - *The exterior angle of a cyclic quadrilateral is equal to the opposite interior angle (without proof).*

3. Geometry

- (a) *Similarity*

Similarity, conditions of similar triangles.

 - (i) *As a size transformation.*

(iii) Tangent and Secant Properties:

- The tangent at any point of a circle and the radius through the point are perpendicular to each other.
- If two circles touch, the point of contact lies on the straight line joining their centres.
- From any point outside a circle, two tangents can be drawn, and they are equal in length.
- If two chords intersect internally or externally then the product of the lengths of the segments are equal.
- If a chord and a tangent intersect externally, then the product of the lengths of segments of the chord is equal to the square of the length of the tangent from the point of contact to the point of intersection.
- If a line touches a circle and from the point of contact, a chord is drawn, the angles between the tangent and the chord are respectively equal to the angles in the corresponding alternate segments.

Note: Proofs of the theorems given above are to be taught unless specified otherwise.

(iv) Constructions

- (a) Construction of tangents to a circle from an external point.
- (b) Circumscribing and inscribing a circle on a triangle and a regular hexagon.

4. Mensuration

Area and volume of solids – Cylinder, Cone and Sphere.

Three-dimensional solids - right circular cylinder, right circular cone and sphere: Area (total surface and curved surface) and Volume. Direct application problems including cost, Inner and Outer volume and melting and recasting method to find the volume or surface area of a new solid. Combination of solids included.

Note: Problems on Frustum are not included.

5. Trigonometry

- (a) Using Identities to solve/prove simple algebraic trigonometric expressions
$$\sin^2 A + \cos^2 A = 1$$
$$1 + \tan^2 A = \sec^2 A$$
$$1 + \cot^2 A = \operatorname{cosec}^2 A; 0 \leq A \leq 90^\circ$$
- (b) Heights and distances: Solving 2-D problems involving angles of elevation and depression using trigonometric tables.

Note: Cases involving more than two right angled triangles excluded.

6. Statistics

Statistics – basic concepts, Mean, Median, Mode. Histograms and Ogive.

(a) Computation of:

- Measures of Central Tendency: Mean, median, mode for raw and arrayed data. Mean*, median class and modal class for grouped data. (both continuous and discontinuous).

* Mean by all 3 methods included:

$$\text{Direct} \quad : \quad \frac{\sum fx}{\sum f}$$

$$\text{Short-cut} \quad : \quad A + \frac{\sum fd}{\sum f} \text{ where } d = x - A$$

$$\text{Step-deviation: } A + \frac{\sum ft}{\sum f} \times i \text{ where } t = \frac{x - A}{i}$$

- (b) Graphical Representation. Histograms and Less than Ogive.

- Finding the mode from the histogram, the upper quartile, lower Quartile and median etc. from the ogive.
- Calculation of inter Quartile range.

7. Probability

Random experiments, Sample space, Events, definition of probability, Simple problems on single events.

SI UNITS, SIGNS, SYMBOLS AND ABBREVIATIONS

(1) Agreed conventions

- Units may be written in full or using the agreed symbols, but no other abbreviation may be used.
- The letter 's' is never added to symbols to indicate the plural form.
- A full stop is not written after symbols for units unless it occurs at the end of a sentence.
- When unit symbols are combined as a quotient, *e.g.*, metre per second, it is recommended that it should be written as m/s, or as m s^{-1} .
- Three decimal signs are in common international use: the full point, the mid-point and the comma. Since the full point is sometimes used for multiplication and the comma for spacing digits in large numbers, it is recommended that the mid-point be used for decimals.

(2) Names and symbols

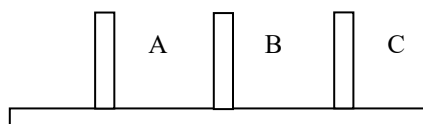
In general			
Implies that	$\Rightarrow$	is logically equivalent to	$\Leftrightarrow$
Identically equal to	$\equiv$	is approximately equal to	$\gg$
In set language			
Belongs to	$\in$	does not belong to	$\notin$
is equivalent to	$\leftrightarrow$	is not equivalent to	$\nleftrightarrow$
union	$\cup$	intersection	$\cap$
universal set	ξ	is contained in	$\subset$
natural (counting) numbers	$\mathbb{N}$	the empty set	$\emptyset$
whole numbers		whole numbers	$\mathbb{W}$
integers	$\mathbb{Z}$	real numbers	$\mathbb{R}$
In measures			
Kilometre	km	Metre	m
Centimetre	cm	Millimetre	mm
Kilogram	kg	Gram	g
Litre	L	Centilitre	cL
square kilometre	km^2	Square meter	m^2
square centimetre	cm^2	Hectare	ha
cubic metre	m^3	Cubic centimetre	cm^3
kilometres per hour	km/h	Metres per second	m/s

INTERNAL ASSESSMENT

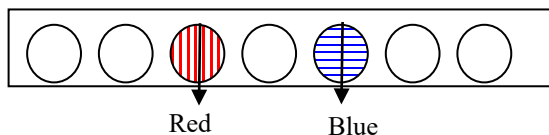
The minimum number of assignments: Two assignments as prescribed by the teacher.

Suggested Assignments

- Comparative newspaper coverage of different items.
- Survey of various types of Bank accounts, rates of interest offered.
- Planning a home budget.
- Conduct a survey in your locality to study the mode of conveyance / Price of various essential commodities / favourite sports. Represent the data using a bar graph / histogram and estimate the mode.
- To use a newspaper to study and report on shares and dividends.
- Set up a dropper with ink in it vertical at a height say 20 cm above a horizontally placed sheet of plain paper. Release one ink drop; observe the pattern, if any, on the paper. Vary the vertical distance and repeat. Discover any pattern of relationship between the vertical height and the ink drop observed.
- You are provided (or you construct a model as shown) - three vertical sticks (size of a pencil) stuck to a horizontal board. You should also have discs of varying sizes with holes (like a doughnut). Start with one disc; place it on (in) stick A. Transfer it to another stick (B or C); this is one move (m). Now try with two discs placed in A such that the large disc is below, and the smaller disc is above (number of discs = $n=2$ now). Now transfer them one at a time in B or C to obtain similar situation (larger disc below). How many moves? Try with more discs ($n = 1, 2, 3$, etc.) and generalise.



- The board has some holes to hold marbles, red on one side and blue on the other. Start with one pair. Interchange the positions by making one move at a time. A marble can jump over another to fill the hole behind. The move (m) equal 3. Try with 2 ($n=2$) and more. Find the relationship between n and m .



- Take a square sheet of paper of side 10 cm. Four small squares are to be cut from the corners of the square sheet and then the paper folded at the cuts to form an open box. What should be the size of the squares cut so that the volume of the open box is maximum?
- Take an open box, four sets of marbles (ensuring that marbles in each set are of the same size) and some water. By placing the marbles and water in the box, attempt to answer the question: do larger marbles or smaller marbles occupy more volume in a given space?
- An eccentric artist says that the best paintings have the same area as their perimeter (numerically). Let us not argue whether such sizes increase the viewer's appreciation, but only try and find what sides (in integers only) a rectangle must have if its area and perimeter are to be equal (Note: there are only two such rectangles).

- Find by construction the centre of a circle, using only a 60-30 setsquare and a pencil.
- Various types of "cryptarithm".

EVALUATION

The assignments/project work are to be evaluated by the subject teacher and by an External Examiner. (The External Examiner may be a teacher nominated by the Head of the school, who could be from the faculty, **but not teaching the subject in the section/class**. For example, a teacher of Mathematics of Class VIII may be deputed to be an External Examiner for Class X, Mathematics projects.)

The Internal Examiner and the External Examiner will assess the assignments independently.

Award of Marks (20 Marks)

Subject Teacher (Internal Examiner) 10 marks

External Examiner 10 marks

The total marks obtained out of 20 are to be sent to CISCE by the Head of the school.

The Head of the school will be responsible for the online entry of marks on CISCE's CAREERS portal by the due date.

	Satisfactory techniques.	in using limited range of techniques.	construction and measurement.	places.	mathematics with a limited ability to use it.	
Grade IV	Exhibits and selects an insignificant problem. Uses some unsuitable techniques.	Uses inappropriate mathematical concepts for the assignment.	Commits many mistakes in computation, construction and measurement.	Presentation made is somewhat disorganized and untidy.	Lack of ability to conclude even with considerable help; assignment contributes to mathematical learning to a certain extent.	1 mark for each criterion
Grade V	Exhibits and selects a completely irrelevant problem. Uses unsuitable techniques.	Not able to use mathematical concepts.	Inaccurate computation, construction and measurement.	Presentation made is completely disorganized, untidy and poor.	Assignment does not contribute to mathematical learning and lacks practical applicability.	0 mark